

MECHANICAL DESIGN PORTFOLIO



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PROJECT - 01

PISTON–CRANK MECHANISM ASSEMBLY



Fig. 1 – Piston Crank Mechanism Assembly

Complete assembly of the piston, piston rings, piston pin, connecting rod, and crankshaft.

SOFTWARE USED

SolidWorks 2023

PROJECT TYPE

3D Part Modeling and Assembly

OBJECTIVE

Designed and developed a 3D piston–crank mechanism assembly using SolidWorks by modeling individual components and assembling them to improve practical mechanical design, CAD modeling, and assembly skills.

COMPONENTS MODELED

- Piston
- Piston Rings
- Piston Pin
- Connecting Rod
- Crankshaft

WORK PERFORMED

- Created 3D models of the piston, piston rings, piston pin, connecting rod, and crankshaft.
- Assembled all components using appropriate assembly mates.
- Verified component alignment and assembly fit.
- Applied part modeling and assembly design techniques to complete the piston–crank mechanism.

CAD FEATURES USED

- Sketch
- Extrude Boss/Base
- Extrude Cut
- Revolve
- Fillet
- Chamfer
- Circular Pattern
- Mirror
- Assembly Mates

SKILLS DEMONSTRATED

- 3D Part Modeling
- Assembly Modeling
- Mechanical Design
- CAD Design
- SolidWorks
- Engineering Drawing Basics

PROJECT GALLERY

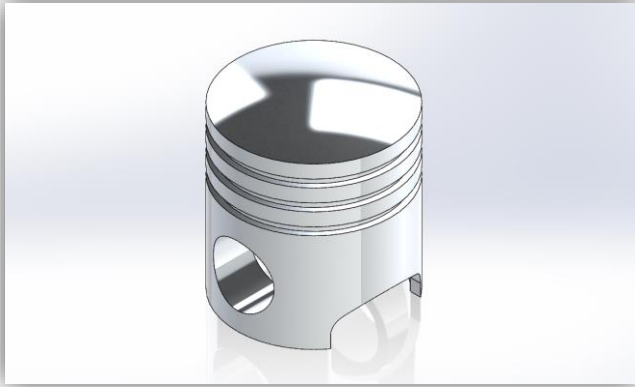


Fig. 2 – Piston

Main reciprocating component of the engine.



Fig. 3 – Connecting Rod

Transmits motion from the piston to the crankshaft.

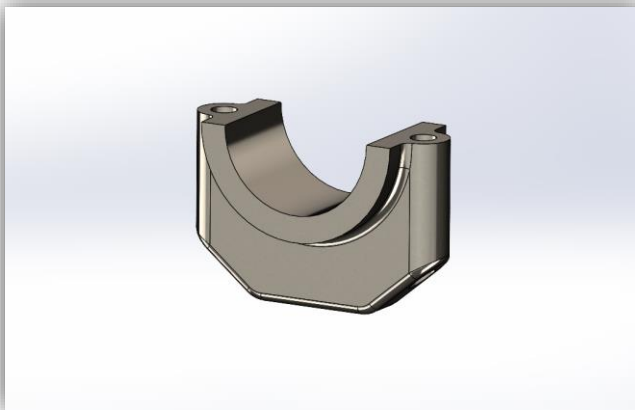


Fig. 3.1 – Connecting Rod Cap

Secures the connecting rod to the crankshaft.

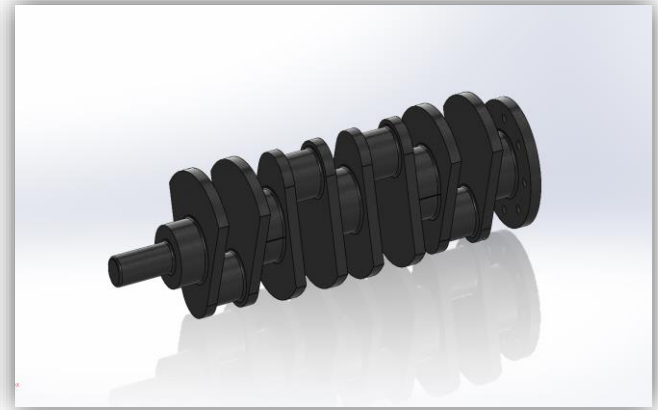


Fig. 4 – Crankshaft

Converts reciprocating motion into rotary motion.

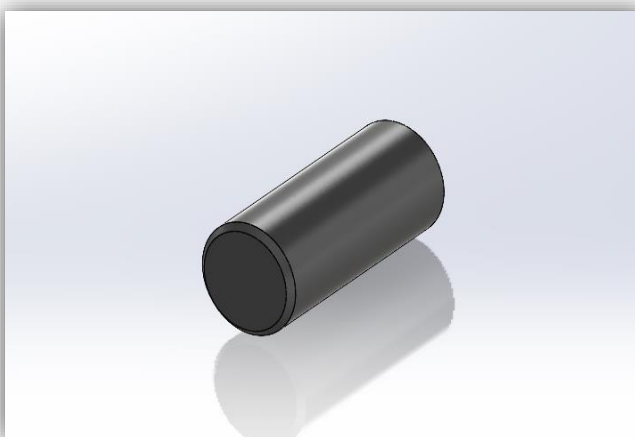


Fig. 5 – Piston Pin (Gudgeon Pin)

Connects the piston to the connecting rod.

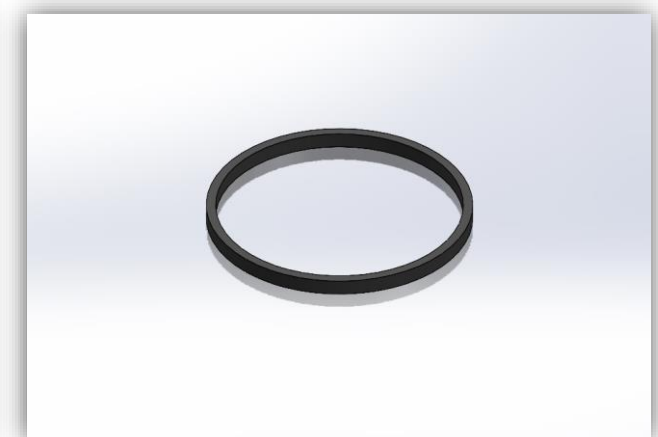


Fig. 6 – Piston Rings

Provide sealing, control oil, and transfer heat.

